Chaeyun Lim 임채윤

Integrated M.S.–Ph.D. Student in Applied Artificial Intelligence

Myongji University, South Korea · DAN Lab.

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RESEARCH INTERESTS

Artificial Intelligence (AI) · Wireless Networking (Wi-Fi MAC) · Extended Reality (XR)

Wi-Fi MAC & Latency – Reinforcement-learning-based contention window control for IEEE 802.11bn to reduce tail latency under dynamic QoS traffic.

Distributed Learning over Ad Hoc Networks – Communication-efficient federated learning (Baton-FL, Chain FL) for UAV and wireless ad hoc environments.

Immersive 3D Media – ROI-aware point cloud downsampling and 3D detection for efficient transmission of XR content.

EDUCATION

2025.03 – Present **Integrated M.S.–Ph.D. in Applied Artificial Intelligence**
Myongji University · DAN Lab.

2019.03 – 2025.02 **B.E. in Information and Communication Engineering (ICE) · Valedictorian**
Myongji University

RESEARCH EXPERIENCE

2025.03 – Present **Integrated M.S.–Ph.D. Student**
Myongji University · DAN Lab.

2024.05 – 2025.02 **Undergraduate Researcher**
Myongji University · DAN Lab.

ACTIVITIES & SERVICE

2024.09 – 2026.02 **XREAL · XR·AI United Society**
9th OB · Dev Team Lead, Team Immersyn (2025.10 – 2026.02)
8th Leader · Head of Dev Team (2025.03 – 2025.08)
7th Junior · Dev Team (2024.09 – 2025.02)

2022.11 – 2024.05 **Military Service** · Republic of Korea Army, Signal (Communications) Soldier

PUBLICATIONS

Journal

[J1] Chaeyun Lim, Seunghyun Park, Hyunhee Park, "**Adaptive Point Cloud Downsampling Based on 3-D ROI Detection for Improved Perceptual Quality**," *IEEE Access*, vol. 15, pp. 68039–68053, 2026. DOI: 10.1109/ACCESS.2026.3689278 **SCIE**

International Conference

[C1] Chaeyun Lim, Hyunhee Park, "**Chain Federated Learning: Communication-Efficient Serverless FedAvg for Wireless Ad Hoc Networks**," The 29th International Conference on Network-Based Information Systems (NBiS 2026), Osaka, Japan, 2026. Oral presentation. DOI: 10.1007/978-3-032-38477-5_3 **SCOPUS** **BEST PAPER**

[C2] Chaeyun Lim, Yongho Kim, Hyunhee Park, "**Adaptive ROI-Aware Point Cloud Downsampling for 3D Media Transmission**," International Conference on Broadband and Wireless Computing, Communication and Applications (BWCCA 2025), UAE (Online), 2025. Oral presentation. DOI: 10.1007/978-3-032-10347-5_6 **SCOPUS**

Domestic Conference (Korea)

[D1] Chaeyun Lim, Hyunhee Park, "**Baton-FL: Distributed Federated Learning via Model Passing for UAV Ad Hoc Networks**," KICS Summer Annual Conference 2026, Jeju, Korea, 2026. Poster presentation.

[D2] Chaeyun Lim, Hyunhee Park, "**Multi-head DQN-Based Adaptive Contention Window Exponent Control for IEEE 802.11bn P95 Latency Reduction in Dynamic QoS Traffic Environments**," Joint Conference on Communications and Information (JCCI 2026), Yeosu, Korea, 2026. Oral presentation.

[D3] Chaeyun Lim, Hyunhee Park, "**Geometry-guided Multi-View Data Augmentation from a Single Image with Generative Completion**," KICS Winter Annual Conference 2026, Yongpyeong, Korea, 2026. Oral presentation. **OUTSTANDING PAPER**

[D4] Chaeyun Lim, Hyunhee Park, "**3D Object Detection based on Multi-Axis Plane Projection for ROI-Preserving Point Cloud Downsampling**," Korea Artificial Intelligence Conference (KAIC 2025), Jeju, Korea, 2025. Oral presentation. **OUTSTANDING PAPER**

[D5] Chaeyun Lim, Hyunhee Park, "**An Idea Proposal for Egocentric 3D Pose Estimation Using Wi-Fi CSI**," Joint Conference on Communications and Information (JCCI 2025), Sokcho, Korea, 2025. Oral presentation.

[D6] Chaeyun Lim, Hyunhee Park, "**CBAM-YOLO11pico: On Smartphones and VR/AR Devices Proposing an Ultra-lightweight Model for Real-Time Object Detection**," KICS Winter Annual Conference 2025, Yongpyeong, Korea, 2025. Undergraduate poster presentation.

[D7] Chaeyun Lim, Sohyeon Jang, Hyunhee Park, "**Development of MR-based AI Companion Contents for Utilizing XR·AI technology in Daily life**," KICS Autumn Annual Conference 2024, Gyeongju, Korea, 2024. Undergraduate poster presentation.

KICS: Korean Institute of Communications and Information Sciences. Domestic conference papers are in Korean.

Patent

[P1] Chaeyun Lim, Hyunhee Park, "**Apparatus and Method for ROI-Based Adaptive Point Cloud Downsampling**," Korean Patent Application No. 10-2026-0085827, filed May 12, 2026 (pending). Myongji University Industry-Academic Cooperation Foundation.

RESEARCH PROJECTS

Funded projects at DAN Lab. Projects are led by the lab's principal investigator unless marked PI.

2026.10 – 2028.09 **Joint Energy-QoE Optimization for Video Delivery in Satellite-UAV-6G Networks via Stochastic Modeling and Generative AI**
Korea–France Cooperation Foundation Program (PHC STAR – Partenariat Hubert Curien) · Joint research with IRISA / Université de Rennes
Role: Main researcher

2026.09 – 2027.08 **Meta-Reinforcement Learning for Adaptive Channel Access Optimization in Self-Healing Multi-AP Networks**
National Research Foundation of Korea (NRF) · Graduate Research Fellowship Program
Role: Principal Investigator · sole awardee of a competitive national research fellowship **PI**

2025.04 – 2029.03 **Development of Hybrid Inspection Equipment Incorporating Deep-Learning-Based Optical Character Recognition (OCR) and Chipping Detection**
Korea Technology and Information Promotion Agency for SMEs (TIPA) · SME Technology Innovation Development Program (Export-Oriented) · Joint research institution
Role: Leading research and experiments

2025.04 – 2025.10 **Development of an Embedded-AI Virtual Keypad for Small Edge Devices**
Industry-commissioned research project · Innopiatech Co., Ltd.
Role: Main researcher for the year · research, development, experiments and final report

2024.08 – 2027.12 **Development of a Vision Recognition Technology SDK Based on Domestic AI Semiconductors**
Ministry of Trade, Industry and Energy (MOTIE) / KEIT · Materials & Components Technology Development Program (Heterogeneous Technology Convergence) · Joint research institution
Role: Leading research, experiments and report writing (since 2025.03)

2022.03 – 2027.02 **AI-MAC Protocol on Distributed Machine Learning for Intelligent Flying Base Stations**
National Research Foundation of Korea (NRF) · Basic Science Research Program (Mid-Career Researcher)
Role: Research and paper writing (since 2025.03)

AWARDS

2026.09 **Best Paper Award**, NBiS 2026 (The 29th International Conference on Network-Based Information Systems)

2026.07 **Excellence Award**, Gyeonggi RISE G7 (AI) Industry-Coupled PBL (IC-PBL+) Competition

2026.02 **Outstanding Paper Award (Haedong Award)**, Korean Institute of Communications and Information Sciences (KICS)

2025.09 **Outstanding Paper Award**, Korea Artificial Intelligence Conference (KAIC 2025)

2025.02 **Valedictorian** (Top of Graduating Class), Dept. of Information and Communication Engineering, Myongji University

2024.10 **Encouragement Award**, XR Developer Challenge (Metaverse Developer Competition)

SCHOLARSHIPS

2026.06 **Merit Scholarship for Master's Students in Science & Engineering**, Korea Student Aid Foundation (KOSAF)

2023.06 **National Science & Technology Scholarship**, Korea Student Aid Foundation (KOSAF)
100% scholarship for 4 semesters (2023 Spring – 2024 Fall)

2019 – 2023 **Baekma Scholarship (Academic Merit Scholarship)**, Myongji University
Awarded 4 times, including two 100% awards

CERTIFICATIONS

2024.06 **Engineer Information Processing** (Korean National Certification)

2024.03 **Pilot of an Ultra Light Vehicle** (Unmanned Multicopter, 1st)